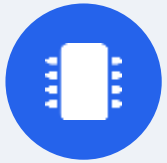
Number System Conversions



Binary Number

- Binary → Decimal
- Binary → Octal
- Binary → Hexadecimal



Octal Number

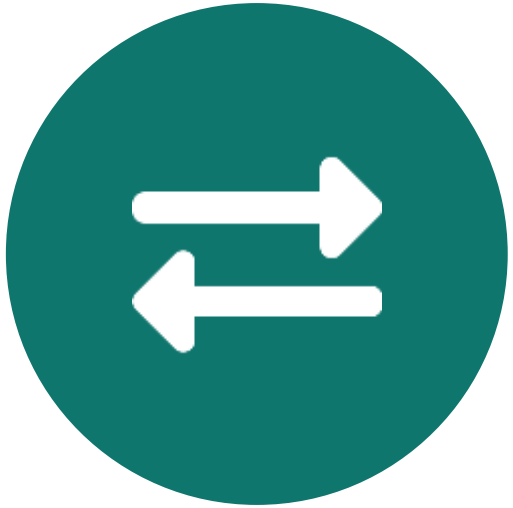
- Octal → Decimal
- Octal → Binary
- Octal → Hexadecimal

BINARY CONVERSIONS

Binary → Decimal

Binary → Octal

Binary → Hexadecimal



Binary Number System

Definition:

Binary is the language of computers. It uses only two digits.

It's used to store, access, and manipulate data in computers.

Digits:

0	1

Base (Radix): 2

Representation:

$(101110111)_2$ $(11001)_2$ $(101101)_2$ $(1101.101)_2$

Binary Place Values

Power	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	.	2^{-1}	2^{-2}	2^{-3}
Value	128	64	32	16	8	4	2	1	.	0.5	0.25	0.125

Moving one place to the left multiplies the value by 2.

Moving one place to the right divides the value by 2.

Binary → Decimal: Method

1. Write the place value (power of 2) under every bit.
2. Multiply each bit by its place value.
3. Add all the results to get the decimal number.

$$\text{Decimal} = \Sigma\left(\text{bit} \times 2^{\text{position}}\right)$$

Binary → Decimal | $(1011)_2$

Without point

Position :	3	2	1	0
Binary :	1	0	1	1

Value:

$$1\times(2)^3 + 0\times(2)^2 + 1\times(2)^1 + 1\times(2)^0$$

$$8 + 0 + 2 + 1 = (11)_{10}$$

Binary → Decimal | $(11001)_2$

Without point

Position :	4	3	2	1	0
Binary :	1	1	0	0	1

Value:

$1\times(2)^4 + 1\times(2)^3 + 0\times(2)^2 + 0\times(2)^1 + 1\times(2)^0$

$$16 + 8 + 0 + 0 + 1 = (25)_{10}$$

Binary → Decimal | $(101101)_2$

Without point

Position :	5	4	3	2	1	0
Binary :	1	0	1	1	0	1

Value:

$1\times(2)^5 + 0\times(2)^4 + 1\times(2)^3 + 1\times(2)^2 + 0\times(2)^1 + 1\times(2)^0$

$$32 + 0 + 8 + 4 + 0 + 1 = (45)_{10}$$

Binary → Decimal | $(101.01)_2$

With point

Position :	2	1	0	.	–1	–2
Binary :	1	0	1	.	0	1

Value:

$1\times(2)^2 + 0\times(2)^1 + 1\times(2)^0 + 0\times(2)^{-1} + 1\times(2)^{-2}$

$$4 + 0 + 1 + 0 + 0.25 = (5.25)_{10}$$

Binary → Decimal | $(110.101)_2$

With point

Position :	2	1	0	.	–1	–2	–3
Binary :	1	1	0	.	1	0	1

Value:

$1\times(2)^2 + 1\times(2)^1 + 0\times(2)^0 + 1\times(2)^{-1} + 0\times(2)^{-2} + 1\times(2)^{-3}$

$$4 + 2 + 0 + 0.5 + 0 + 0.125 = (6.625)_{10}$$

Binary → Decimal | $(1101.11)_2$

With point

Position :	3	2	1	0	.	–1	–2
Binary :	1	1	0	1	.	1	1

Value:

$$1\times(2)^3 + 1\times(2)^2 + 0\times(2)^1 + 1\times(2)^0 + 1\times(2)^{-1} + 1\times(2)^{-2}$$

$$8 + 4 + 0 + 1 + 0.5 + 0.25 = (13.75)_{10}$$

Why does Octal use 3 bits?

Octal has 8 digits (base 8):

0	1	2	3	4	5	6	7

2 bits

2×2

= 4 patterns

Not enough for 0 to 7

3 bits

$2 \times 2 \times 2$

= 8 patterns

Exactly 8 digits: 0 to 7

4 bits

$2 \times 2 \times 2 \times 2$

= 16 patterns

More than needed

Base 8 = 2^3 → 1 octal digit = 3 bits

How to make the 3-bit table

$2^2 = 4$	$2^1 = 2$	$2^0 = 1$	Digit
0	0	0	0
0	0	1	1
0	1	0	2
0	1	1	3
1	0	0	4
1	0	1	5
1	1	0	6
1	1	1	7

2^0 column

0 1 0 1 0 1 0 1

changes in every row

2^1 column

0 0 1 1 0 0 1 1

changes after every 2 rows

2^2 column

0 0 0 0 1 1 1 1

changes after every 4 rows

Binary → Octal: Method

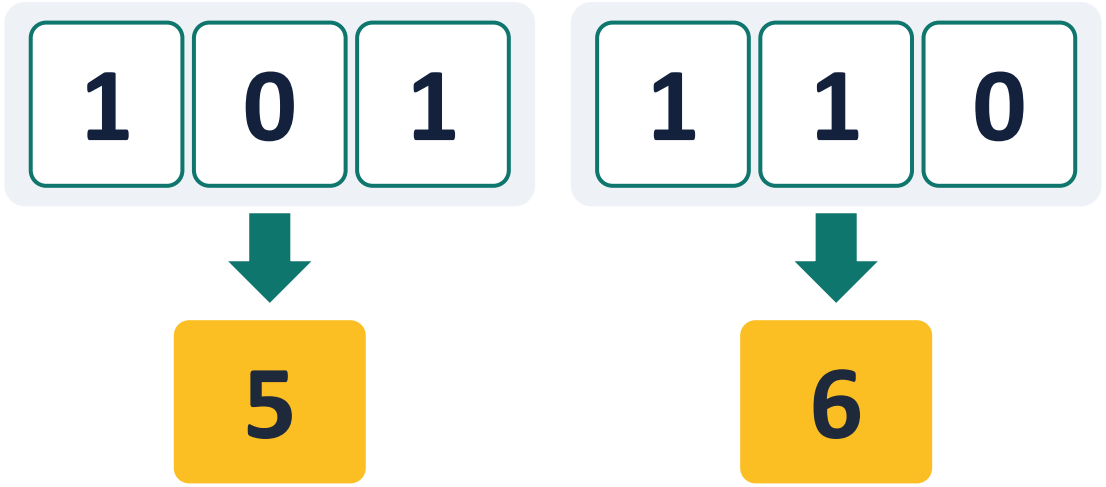
1. Make groups of 3 bits, starting from the binary point.
2. Complete the last group with 0s (left of integer, right of fraction).
3. Replace each group with its octal digit.

Lookup table: 3 bits → octal digit

000	001	010	011	100	101	110	111
0	**1**	**2**	**3**	**4**	**5**	**6**	**7**

Binary → Octal | $(101110)_2$

Without point



3-bit table (the values used in this example are highlighted)

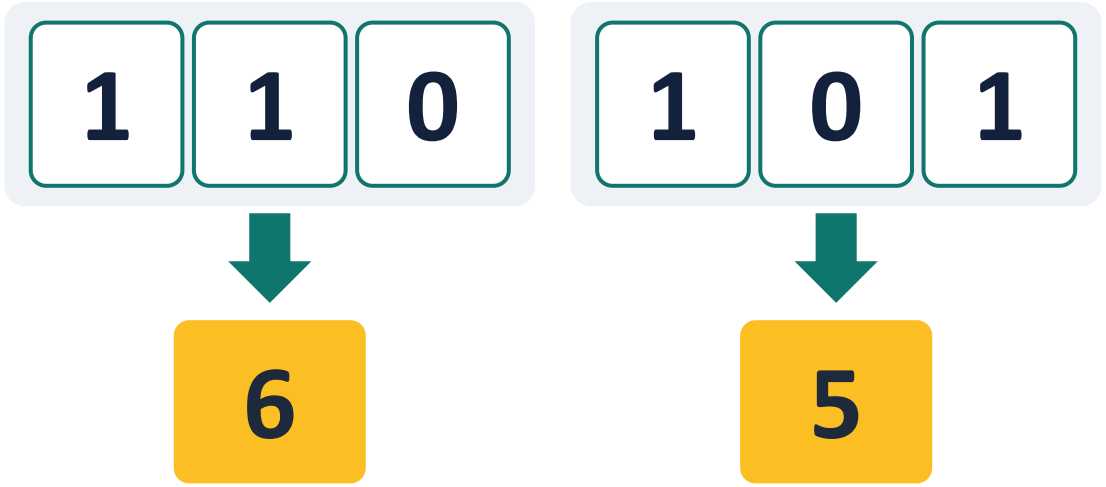
000	001	010	011	100	101	110	111
0	1	2	3	4	5	6	7

Groups of 3 bits.

Answer: $(56)_8$

Binary → Octal | $(110101)_2$

Without point



3-bit table (the values used in this example are highlighted)

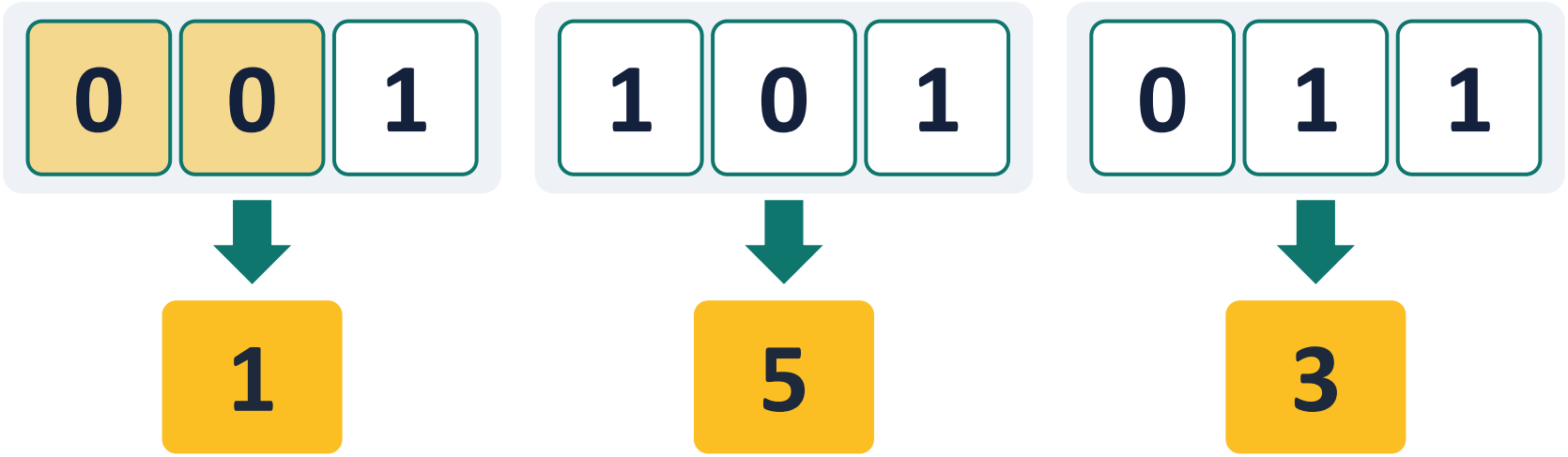
000	001	010	011	100	101	110	111
0	**1**	**2**	**3**	**4**	**5**	**6**	**7**

Groups of 3 bits.

Answer: $(65)_8$

Binary → Octal | $(1101011)_2$

Without point



3-bit table (the values used in this example are highlighted)

000	001	010	011	100	101	110	111
0	1	2	3	4	5	6	7

Groups of 3 bits. Amber 0s are added to complete a group.

Answer: $(153)_8$

Binary → Octal | $(101.11)_2$

With point



3-bit table (the values used in this example are highlighted)

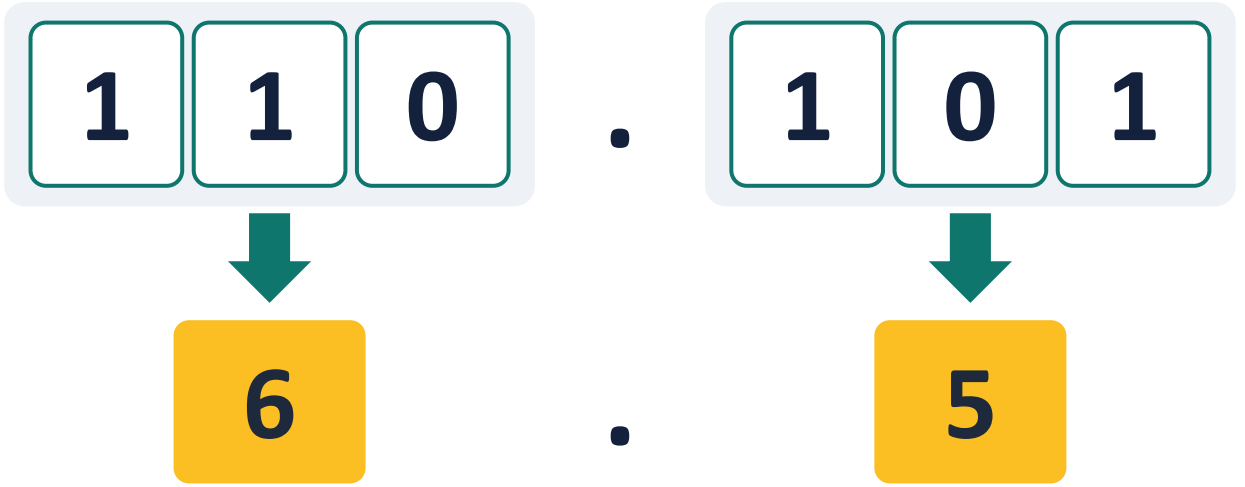
000	001	010	011	100	101	110	111
0	1	2	3	4	5	6	7

Groups of 3 bits. Amber 0s are added to complete a group.

Answer: $(5.6)_8$

Binary → Octal | $(110.101)_2$

With point



3-bit table (the values used in this example are highlighted)

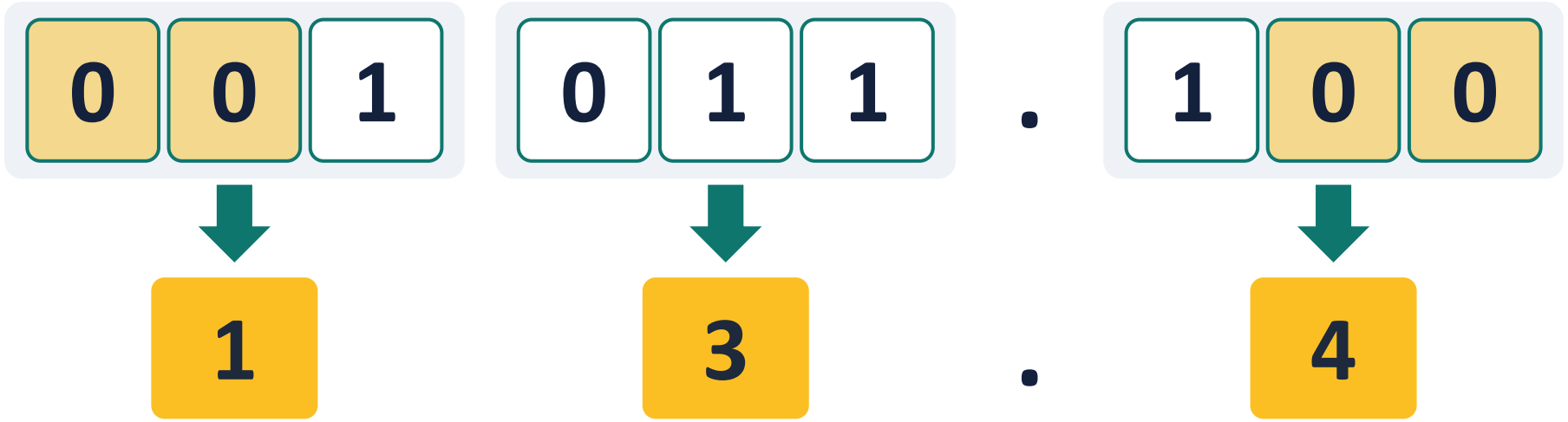
000	001	010	011	100	101	110	111
0	**1**	**2**	**3**	**4**	**5**	**6**	**7**

Groups of 3 bits.

Answer: $(6.5)_8$

Binary → Octal | $(1011.1)_2$

With point



3-bit table (the values used in this example are highlighted)

000	001	010	011	100	101	110	111
0	1	2	3	4	5	6	7

Groups of 3 bits. Amber 0s are added to complete a group.

Answer: $(13.4)_8$

Why does Hexadecimal use 4 bits?

Hexadecimal has 16 digits (base 16):

0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F

3 bits

$2 \times 2 \times 2$

= 8 patterns

Not enough for 0 to F

4 bits

$2 \times 2 \times 2 \times 2$

= 16 patterns

Exactly 16 digits: 0 to F

5 bits

$2 \times 2 \times 2 \times 2 \times 2$

= 32 patterns

More than needed

Base 16 = 2^4 → 1 hex digit = 4 bits

How to make the 4-bit table

$2^3 = 8$	$2^2 = 4$	$2^1 = 2$	$2^0 = 1$	Digit
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7

$2^3 = 8$	$2^2 = 4$	$2^1 = 2$	$2^0 = 1$	Digit
1	0	0	0	8
1	0	0	1	9
1	0	1	0	10 = A
1	0	1	1	11 = B
1	1	0	0	12 = C
1	1	0	1	13 = D
1	1	1	0	14 = E
1	1	1	1	15 = F

Start from 0 and keep adding 1. The right bit changes every row, the next every 2 rows, then every 4 rows, and the left bit every 8 rows. Digits 10 to 15 are written as A to F.

Binary → Hexadecimal: Method

1. Make groups of 4 bits, starting from the binary point.
2. Complete the last group with 0s (left of integer, right of fraction).
3. Replace each group with its hex digit (A = 10 … F = 15).

Lookup table: 4 bits → hex digit (= decimal value)

0000 **0** = 0	0001 **1** = 1	0010 **2** = 2	0011 **3** = 3	0100 **4** = 4	0101 **5** = 5	0110 **6** = 6	0111 **7** = 7
1000 **8** = 8	1001 **9** = 9	1010 **A** = 10	1011 **B** = 11	1100 **C** = 12	1101 **D** = 13	1110 **E** = 14	1111 **F** = 15

Binary → Hexadecimal | $(11010111)_2$

Without point



4-bit table (the values used in this example are highlighted)

0000 **0**	0001 **1**	0010 **2**	0011 **3**	0100 **4**	0101 **5**	0110 **6**	0111 **7**
1000 **8**	1001 **9**	1010 **A** = 10	1011 **B** = 11	1100 **C** = 12	1101 **D** = 13	1110 **E** = 14	1111 **F** = 15

Groups of 4 bits.

Answer: $(D7)_{16}$

Binary → Hexadecimal | $(101101)_2$

Without point



4-bit table (the values used in this example are highlighted)

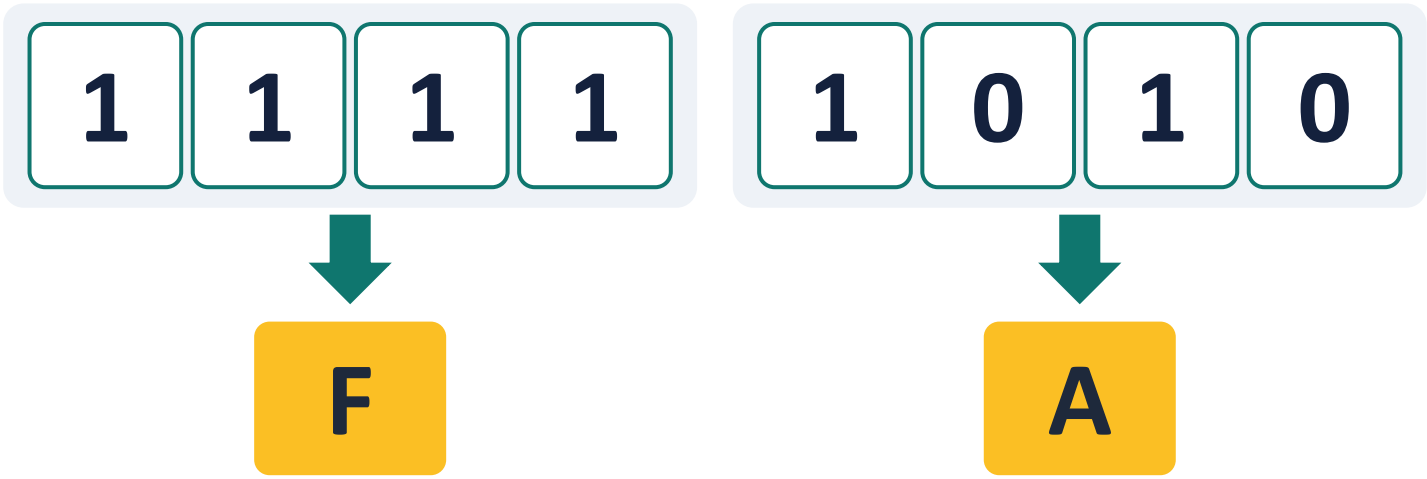
0000 **0**	0001 **1**	0010 **2**	0011 **3**	0100 **4**	0101 **5**	0110 **6**	0111 **7**
1000 **8**	1001 **9**	1010 **A** = 10	1011 **B** = 11	1100 **C** = 12	1101 **D** = 13	1110 **E** = 14	1111 **F** = 15

Groups of 4 bits. Amber 0s are added to complete a group.

Answer: $(2D)_{16}$

Binary → Hexadecimal | $(11111010)_2$

Without point



4-bit table (the values used in this example are highlighted)

0000 **0**	0001 **1**	0010 **2**	0011 **3**	0100 **4**	0101 **5**	0110 **6**	0111 **7**
1000 **8**	1001 **9**	1010 **A** = 10	1011 **B** = 11	1100 **C** = 12	1101 **D** = 13	1110 **E** = 14	1111 **F** = 15

Groups of 4 bits.

Answer: $(FA)_{16}$

Binary → Hexadecimal | $(1011.1)_2$

With point



4-bit table (the values used in this example are highlighted)

0000 **0**	0001 **1**	0010 **2**	0011 **3**	0100 **4**	0101 **5**	0110 **6**	0111 **7**
1000 **8**	1001 **9**	1010 **A** = 10	1011 **B** = 11	1100 **C** = 12	1101 **D** = 13	1110 **E** = 14	1111 **F** = 15

Groups of 4 bits. Amber 0s are added to complete a group.

Answer: $(B.8)_{16}$

Binary → Hexadecimal | $(110.101)_2$

With point



4-bit table (the values used in this example are highlighted)

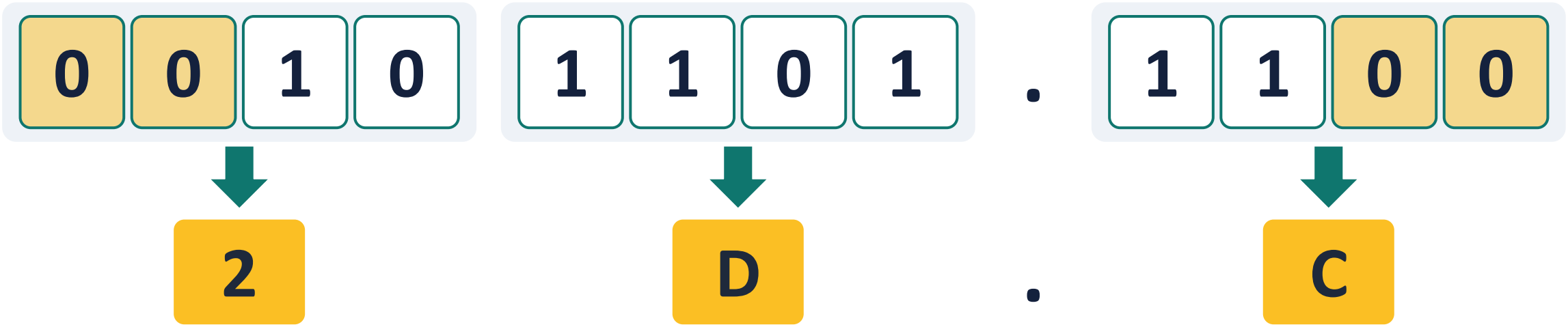
0000 **0**	0001 **1**	0010 **2**	0011 **3**	0100 **4**	0101 **5**	0110 **6**	0111 **7**
1000 **8**	1001 **9**	1010 **A** = 10	1011 **B** = 11	1100 **C** = 12	1101 **D** = 13	1110 **E** = 14	1111 **F** = 15

Groups of 4 bits. Amber 0s are added to complete a group.

Answer: $(6.A)_{16}$

Binary → Hexadecimal | $(101101.11)_2$

With point



4-bit table (the values used in this example are highlighted)

0000 **0**	0001 **1**	0010 **2**	0011 **3**	0100 **4**	0101 **5**	0110 **6**	0111 **7**
1000 **8**	1001 **9**	1010 **A** = 10	1011 **B** = 11	1100 **C** = 12	1101 **D** = 13	1110 **E** = 14	1111 **F** = 15

Groups of 4 bits. Amber 0s are added to complete a group.

Answer: $(2D.C)_{16}$